PEF 6000 - Special topics of dynamics of structures
Module 2 - Introduction to random vibrations

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Outline

Outline

Objectives and references

- To introduce to basic aspects related to random vibrations;
- Focuses of the classes: Statistics, Fourier Series/Transform, frequency domain analysis and relation between statistics;
- Examples of references

1. Thomson, W.T. & Dahleh, M.D., 2005. *Theory of vibration with application*. Pearson education.
2. Meirovitch, L., 2000. *Fundamentals of vibrations*. McGraw-Hill.
3. Aguirre, L.A., 2015. *Introdução à identificação de sistemas técnicas lineares e não lineares: Teoria e Aplicação.* Editora UFMG.

Outline

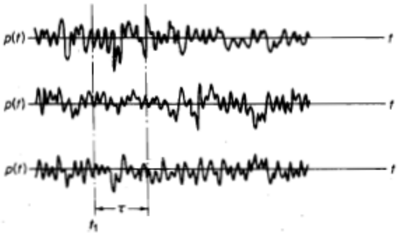
Introduction

- Up to this point, focus has been placed on the deterministic dynamics, i.e., both the structural properties and the external excitation are fully known;
- In technological applications, a number of dynamic phenomena are non-deterministic (or, in other words, exhibit a random behavior). Examples: seismic excitation, loads due to surface waves in offshore structures, aerodynamic loads associated with turbulence...
- The focus of this class is on problems characterized by random excitation, but with deterministic structural properties. Despite interesting, random vibrations in which the structural properties are defined by random variables are out of the scope of this class.

Introduction

- Sample: In this class, is a time-history of a certain quantity. For example, the time-history of acceleration of a certain point of a floating unit during one day. It is a random variable;
- Ensemble: A collection of samples. It is a random (or stochastic) process.

(a) Sample.



(b) Ensemble.

Extracted from Thomson & Daleh (2005).

Basic statistics - Ensemble averages

Consider that we have an ensemble with N_s samples.

- The mean value at an instant t_k is

$$\mu_x(t_k) = \lim_{N_s \to \infty} \frac{1}{N_s} \sum_{i=1}^{N_s} x_i(t_k) \tag{1}$$

- The autocorrelation function is

$$R(t_k, t_k + \tau) = \lim_{N_s \to \infty} \frac{1}{N_s} \sum_{i=1}^{N_s} x_i(t_k) x_i(t_k + \tau) \tag{2}$$

- If both $\mu(t_k)$ and $R(t_k, t_k + \tau)$ depend on t_k: nonstationary random process;
- If neither $\mu(t_k)$ nor $R(t_k, t_k + \tau)$ depend on t_k: weakly stationary random process;
- Weakly stationary process: $\mu(t_k) = \mu$ and $R(t_k, t_k + \tau) = R(\tau)$;
- If all the ensemble averages (including higher-order ones not herein presented) are independent of t_k: Strongly stationary process.
- Usually, we assume that if a process is weakly stationary, it is also strongly stationary. We will employed just the nomenclature "stationary".

Basic statistics - time averages

- Time-averaging:

$$\mu_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)dt \tag{3}$$

- In practical applications, the time-histories are given in the form of a vector of finite size, containing the values of $x(t)$ at some time instants (usually employing a constant time step). In this case

$$\mu_x = \frac{1}{N_t} \sum_{k=1}^{N_t} x_k \tag{4}$$

  N_t being the size of the time history.
- Mean-square value: Average value of $x^2(t)$.

$$\psi_x^2 = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)dt \tag{5}$$

- Temporal correlation between two signals:

$$R(x_1, x_2, \tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x_1(t)x_2(t+\tau)dt \tag{6}$$

Basic statistics

- Auto-correlation function of a real signal:

$$R(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t+\tau)dt \tag{7}$$

- Variance: Average value of $(x(t) - \mu_x)^2$

$$\sigma_x^2 = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} (x(t)-\mu_x)^2 dt = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} (x^2(t) - 2x(t)\mu_x + \mu_x^2)dt = \psi_x^2 - \mu_x^2 \tag{8}$$

- Standard deviation: positive root of σ_x^2;
- Root-mean square (r.m.s): Square-root of the mean-square value. If the signal has zero average value, the r.m.s matches the standard deviation;
- If a process is stationary and the time averages are the same for each sample, the process is ergodic.
- An ergodic process is stationary. A stationary process can not be ergodic.

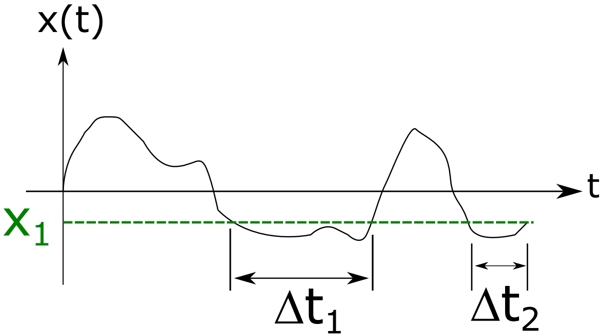
Other definitions

Definitions shown in Aguirre (2015)

- “Um processo estocástico é estacionário no sentido estrito se sua densidade de probabilidade se mantém inalterada após a mudança na origem no tempo”;
- “Um processo estocástico é estacionário no sentido amplo (ou estacionaridade fraca) se sua média é constante';'
- “Se um processo for estacionário no sentido amplo, então sua função de correlação somente dependerá da diferença temporal considerada $|t_1 - t_2| = \tau$”;

Probability theory

- For random variables, we are interested in defining the probability of this variable assuming values larger (or smaller) than a certain value.



- Based on the above figure, calculate $P[x < x_1]$.
- Answer:

$$P[x < x_1] = P[x_1] = \lim_{t \to \infty} \frac{1}{t} \sum \Delta t_k \tag{9}$$

Probability theory

- Notice that

$$P[x < x_1] = P[x_1] = \begin{cases} 1, & x_1 \to \infty \\ 0, & x_1 \to -\infty \end{cases} \tag{10}$$

- $P[x_1]$ is the cumulative probability distribution function.
- Now, we want to evaluate the probability of $x(t)$ lying between x_1 and $x_1 + \Delta x$. This is done by computing:

$$P[x_1 + \Delta x] - P[x_1] \tag{11}$$

- We define the density probability function as:

$$p(x) = \lim_{\Delta x \to 0} \frac{P[x_1 + \Delta x] - P[x_1]}{\Delta x} = \frac{dP}{dx} \tag{12}$$

- Hence, we can write $P[x_1]$ as function of $p(x)$ by computing the integral

$$P[x_1] = \int_{-\infty}^{x_1} p(x)dx \tag{13}$$

Probability theory

- Notice that $P[\infty] = 1$ (area below the curve defined by $p(x)$)
- Mean value (Centroid of the area below the curve):

$$\bar{x} = \int_{-\infty}^{\infty} x p(x) dx \tag{14}$$

- Variance (Moment of inertia around the axis defined by $\bar{x}$)

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 p(x) dx \tag{15}$$

- Quadratic mean value (Moment of inertia around the axis defined by the ordinate axis)

$$\overline{x^2} = \int_{-\infty}^{\infty} x^2 p(x) dx \tag{16}$$

Two important probability density functions

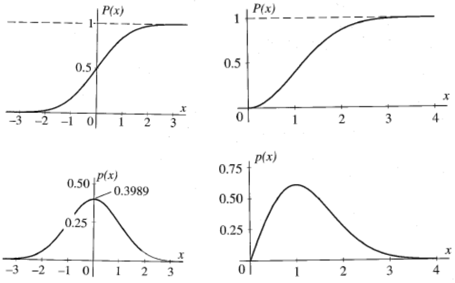
- Gaussian or normal for a random variable of null mean:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} exp(-0.5(x/\sigma)^2) \tag{17}$$

- If we have a random variable with positive values (for example, the amplitude A), Rayleigh's distribution is commonly found:

$$p(A) = \frac{A}{\sigma^2} exp(-0.5(A/\sigma)^2) \tag{18}$$

Two important probability density functions



(a) Normal distribution.

(b) Rayleigh distribution

Extracted from Meirovitch (2000).

Outline

Definitions

- $x(t)$ is said to be periodic of period T if $x(t) = x(t+T), \forall t$;
- Let $x(t)$ be a periodic signal of period $T = 2\pi/\omega$. The Fourier series of this signal is the projection of $x(t)$ onto the set of harmonic function $[\cos n\omega t, \sin n\omega t], n = 0, 1, 2, 3 \ldots, \infty$;
- Considering the usual inner product between two functions $f(t)$ and $g(t)$ $< f, g >= \int_0^T f(t)g(t)dt$ it is easy to notice that the trigonometric functions form a orthogonal basis;

Fourier series - Trigonometric form

- Hence

$$x(t) = \sum_{n=0}^{\infty} \left[a_n \sin(n\omega t) + b_n \cos(n\omega t)\right] =$$
$$= b_0 + \sum_{n=1}^{\infty} \left[a_n \sin(n\omega t) + b_n \cos(n\omega t)\right] \tag{19}$$

with

$$b_n = \frac{\int_0^T x(t)\cos(n\omega t)dt}{\int_0^T \cos^2(n\omega t)dt} = \frac{2}{T}\int_0^T x(t)\cos(n\omega t)dt, n = 1, 2, \ldots \tag{20}$$
$$a_n = \frac{\int_0^T x(t)\sin(n\omega t)dt}{\int_0^T \sin^2(n\omega t)dt} = \frac{2}{T}\int_0^T x(t)\sin(n\omega t)dt, n = 1, 2, \ldots \tag{21}$$
$$b_0 = \frac{\int_0^T x(t)\cos(0\omega t)dt}{\int_0^T \cos^2(0\omega t)dt} = \frac{1}{T}\int_0^T x(t)dt \tag{22}$$

- **Notice that b_0 is the mean value of the signal.**

Fourier series - exponential form

- The Fourier series can be written either in the trigonometric or in the exponential form. In the latter case, we recall the Euler's formula $e^{\imath\theta} = \cos\theta + \imath\sin\theta$ and the following identities:

$$\cos(n\omega t) = \frac{e^{\imath n\omega t} + e^{-\imath n\omega t}}{2} \tag{23}$$

$$\sin(n\omega t) = \frac{e^{\imath n\omega t} - e^{-\imath n\omega t}}{2\imath} = \frac{\imath(e^{-\imath n\omega t} - e^{\imath n\omega t})}{2} \tag{24}$$

- Substituting the above identities in Eq. 19, we have

$$x(t) = b_0 + \sum_{n=1}^{\infty}\left[a_n\left(\frac{\imath(e^{-\imath n\omega t} - e^{\imath n\omega t})}{2}\right) + b_n\left(\frac{e^{\imath n\omega t} + e^{-\imath n\omega t}}{2}\right)\right] =$$

$$= b_0 + \sum_{n=1}^{\infty}\left[\left(\frac{b_n - \imath a_n}{2}\right)e^{\imath n\omega t} + \left(\frac{b_n + \imath a_n}{2}\right)e^{-\imath n\omega t}\right] =$$

$$= b_0 + \sum_{n=-\infty, n\neq 0}^{\infty} \underbrace{\left(\frac{b_n - \imath a_n}{2}\right)}_{c_n} e^{\imath n\omega t} \tag{25}$$

Fourier series - exponential form

- Using the definitions of a_n and b_n in Eq. 25, we obtain

$$c_n = \left(\frac{b_n - \mathbb{i} a_n}{2}\right) = \frac{1}{2}\left(\frac{2}{T}\int_0^T x(t)\cos(n\omega t)dt - \mathbb{i}\frac{2}{T}\int_0^T x(t)\sin(n\omega t)dt\right) =$$
$$= \frac{1}{T}\int_0^T x(t)(\cos(n\omega t) - \mathbb{i}\sin(n\omega t))dt = \frac{1}{T}\int_0^T x(t)e^{-\mathbb{i}n\omega t}dt \qquad (26)$$

Fourier series - example

Consider the periodic function of period T_p given by:

$$p(t) = p_0 t, T_p k < t < T_p(2k+1)/2, k = 0, 1, 2 \ldots \quad (27)$$

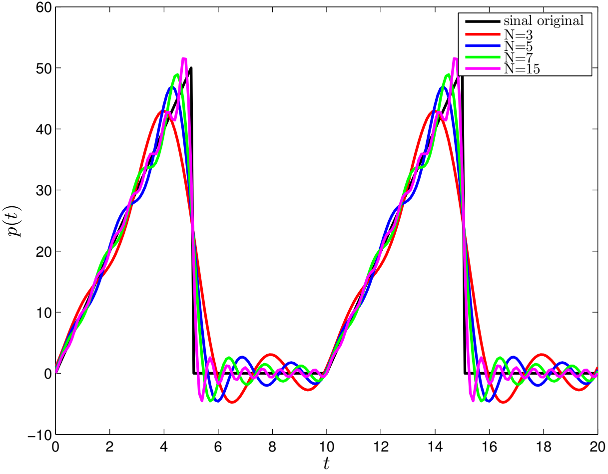
$$p(t) = 0, T_p(2k+1)/2 < t < T_p(k+1), k = 0, 1, 2, \ldots \quad (28)$$

Fourier series - example

- Fundamental frequency $\bar{\omega} = \frac{2\pi}{T_p}$
- The series is truncated in N terms
- $p(t) = b_0 + \sum_{n=1}^{N} (b_n \cos(n\bar{\omega}t) + a_n \sin(n\bar{\omega}t))$
- $b_0 = \frac{1}{T_p} \int_0^{T_p} p(t)dt$
- $b_n = \frac{2}{T_p} \int_0^{T_p} p(t) \cos(n\bar{\omega}t)$
- $a_n = \frac{2}{T_p} \int_0^{T_p} p(t) \sin(n\bar{\omega}t)$

Fourier series - example

Considering $p_0 = 10$ and $T_p = 10$, we have:



Outline

Definitions

- Periodic signals can be studied using the Fourier series. What can we do if the signal is not periodic?
- We consider a non-periodic signal as a periodic one with $T \to \infty$.

Fourier Transform

- In the complex Fourier series, the frequency discretization (i.e., the interval between two identified frequencies) is ω. Here, this interval is defined as $\Delta\omega = \frac{2\pi}{T}$
- Hence

$$x(t) = \lim_{T\to\infty} \sum_{n=-\infty}^{\infty} \frac{1}{T} \left(\int_0^T x(t) e^{-\mathbb{i} n \Delta\omega t} dt \right) e^{in\Delta\omega t} \tag{29}$$

- When $T \to \infty$, the interval between frequencies goes to $d\omega$. In turn, $n\Delta\omega$ goes to ω as $T \to \infty$. Following, we obtain:

$$x(t) = \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-\mathbb{i}\omega t} dt \right) e^{\mathbb{i}\omega t} d\omega \tag{30}$$

Fourier Transform

We define

- Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-\mathbb{i}\omega t} dt \tag{31}$$

- Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{\mathbb{i}\omega t} d\omega \tag{32}$$

- Fourier transform exists if $\int_{-\infty}^{\infty} |x(t)| dt < \infty$, if the number of discontinuities and extrema in $x(t)$ are finite and if the discontinuities are also finite.

Important aspects of the Fourier Transform

- Problem: Consider a 1-dof oscillator of mass m, linear damping constant c and stiffness k. For a known $p(t)$, what is the Fourier Transform of the displacement time history $u(t)$?
- Equation of motion:

$$\ddot{u} + 2\xi\omega\dot{u} + \omega^2 u = \frac{p(t)}{m} \tag{33}$$

$$\omega = \sqrt{\frac{k}{m}}, \xi = \frac{c}{2m\omega} \tag{34}$$

- Answer: Firstly, we consider $p(t) = e^{\mathbb{i}\bar{\omega}t}$. Steady-state responses are characterized by:

$$u(t) = He^{\mathbb{i}\bar{\omega}t} \tag{35}$$

$$\dot{u}(t) = \mathbb{i}\bar{\omega}He^{\mathbb{i}\bar{\omega}t} \tag{36}$$

$$\ddot{u}(t) = -\bar{\omega}^2 He^{\mathbb{i}\bar{\omega}t} \tag{37}$$

- $H = H(\bar{\omega})$ is the complex frequency response function.

Important aspects of the Fourier Transform

- Substituting into the equation of motion:

$$(-\bar{\omega}^2 + \omega^2 + \mathrm{i}2\zeta\omega\bar{\omega})H(\bar{\omega})e^{\mathrm{i}\bar{\omega}t} = \frac{e^{\mathrm{i}\bar{\omega}t}}{m} \tag{38}$$

$$H(\bar{\omega}) = \frac{1}{k\left[\left(1-\left(\frac{\bar{\omega}}{\omega}\right)^2\right) + \mathrm{i}2\zeta\frac{\bar{\omega}}{\omega}\right]} \tag{39}$$

- Now, consider

$$p(t) = \sum_{n=-\infty}^{\infty} P_n e^{\mathrm{i}n\bar{\omega}t} \tag{40}$$

$$P_n = P(n\bar{\omega}) \tag{41}$$

- Owing to the linear character of the mathematical model:

$$u(t) = \sum_{n=-\infty}^{\infty} P(n\bar{\omega})H(n\bar{\omega})e^{\mathrm{i}n\bar{\omega}t} \tag{42}$$

Important aspects of the Fourier Transform

- If the fundamental period $\bar{T} = \frac{2\pi}{\bar{\omega}} \rightarrow \infty$ and using the same approach employed for deriving the Fourier Transform, we obtain:

$$u(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \underbrace{H(\bar{\omega})P(\bar{\omega})}_{U(\bar{\omega})} e^{\mathring{\imath}\bar{\omega}t} d\bar{\omega} \quad (43)$$

- Hence, the Fourier Transform of $u(t)$ is $U(\bar{\omega}) = H(\bar{\omega})P(\bar{\omega})$.
- Another aspect is now discussed. Consider Duhamel's integral:

$$u(t) = \int_{-\infty}^{t} p(\tau)h(t-\tau)d\tau \quad (44)$$

- We consider $\zeta = t - \tau \rightarrow \tau = t - \zeta$. If $\tau = -\infty$, $\zeta = \infty$. If $\tau = t$, $\zeta = 0$. Hence,

$$u(t) = \int_{0}^{\infty} p(t-\zeta)h(\zeta)d\zeta \quad (45)$$

Important aspects of the Fourier Transform

- If $p(t) = e^{\mathbb{i}\omega t}$

$$u(t) = \int_0^\infty e^{\mathbb{i}\omega(t-\zeta)} h(\zeta) d\zeta = e^{\mathbb{i}\omega t} \underbrace{\int_0^\infty h(\zeta) e^{-\mathbb{i}\omega\zeta} d\zeta}_{H(\omega) \mathrm{if} h(t)=0, t<0} \quad (46)$$

- Conclusion: $H(\omega)$ is the Fourier Transform of the impulsive response.
- Note: $u(t) = p(t) * h(t)$ (convolution integral). "The Fourier Transform of the temporal convolution corresponds to the product in the frequency domain".

Properties of the Fourier Transform

- We rewrite the pair of Fourier Transform:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-\mathbb{i}\omega t} dt \tag{47}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{\mathbb{i}\omega t} d\omega \tag{48}$$

- Fourier Transform of $\dot{x}(t)$:

$$\dot{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbb{i}\omega X(\omega) e^{\mathbb{i}\omega t} d\omega \tag{49}$$

  Hence $FT[\dot{x}] = \mathbb{i}\omega FT[x]$.

- If we would like to write $X(f)$ (f in Hz) instead of $X(\omega)$ (ω in rad/s), we recall that $\omega = 2\pi f$ and, then,

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-\mathbb{i}2\pi f t} dt \tag{50}$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{\mathbb{i}2\pi f t} df \tag{51}$$

Parseval's theorem

- For two real signals $x_1(t)$ and $x_2(t)$

$$\int_{-\infty}^{\infty} x_1(t)x_2(t)dt = \int_{-\infty}^{\infty} X_1^*(f)X_2(f)df \tag{52}$$

- Dem:

$$\begin{aligned}
&\int_{-\infty}^{\infty} x_1(t)x_2(t)dt = \int_{-\infty}^{\infty} x_1(t)\left[\int_{-\infty}^{\infty} X_2(f)e^{\imath 2\pi f t}df\right]dt = \\
&\int_{-\infty}^{\infty} X_2(f)\left[\int_{-\infty}^{\infty} x_1(t)e^{\imath 2\pi f t}dt\right]df = \int_{-\infty}^{\infty} X_1^*(f)X_2(f)df = \\
&= \int_{-\infty}^{\infty} X_1(f)X_2^*(f)df
\end{aligned} \tag{53}$$

Outline

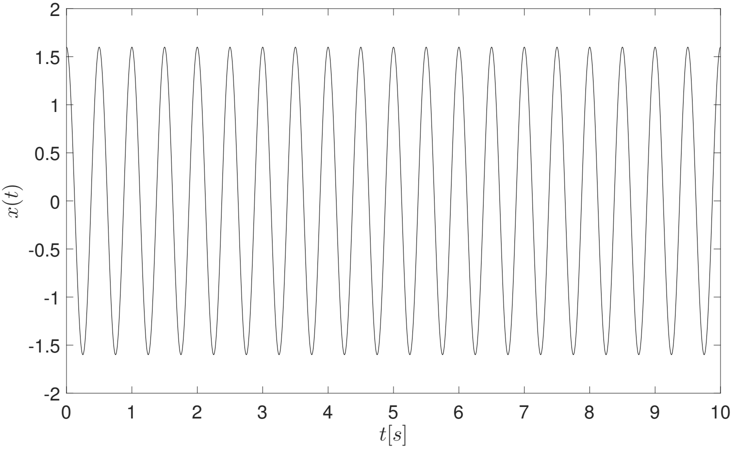
Discrete Fourier Transform

- In practical applications (experiments or numerical simulations), time is discretized (sampled) and, hence, the signals (for example, acceleration of a certain point) are given in the form of a vector of finite size;
- Focus of the class: Cases in which time is given by a vector with N points (size of the sample), sampled at constant frequency (rate) f_s (sampling frequency). The sample period is $T_s = 1/f_s$.
- We need to use the Discrete Fourier Transform (DFT). Fast Fourier Transform (FFT) is an efficient algorithm for calculating the DFT.

Practical aspects

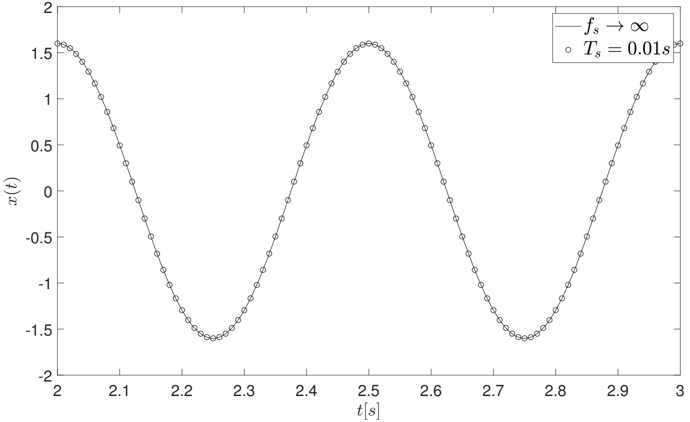
- Now, we discuss how to obtain the amplitude spectrum $X(f)$ from $x(t)$ using `MATLAB`®/`Octave` (and also in `Julia`).
- From the physical point of view, $X(f)$ illustrates the amplitude of each frequency component of $x(t)$;
- Since time is given in discrete form, the amplitude spectrum is also obtained at discrete values of frequency;
- For the sake of illustration, consider the signal given for continuum time $x(t) = A\cos(\omega t)$, with $A = 1.6$ e $\omega = 4\pi \mathrm{rad/s}$

"Continuum" time signal



Practical aspects - sampling

- Now, we consider the discrete signal x_d, obtained from an experiment with constant sample period $T_s = 0.01\text{s}$.



Step 1

Firstly, we obtain the complex amplitude components by means of the DFT. Following, we define the vector with the frequencies in which the amplitude spectrum will be defined.

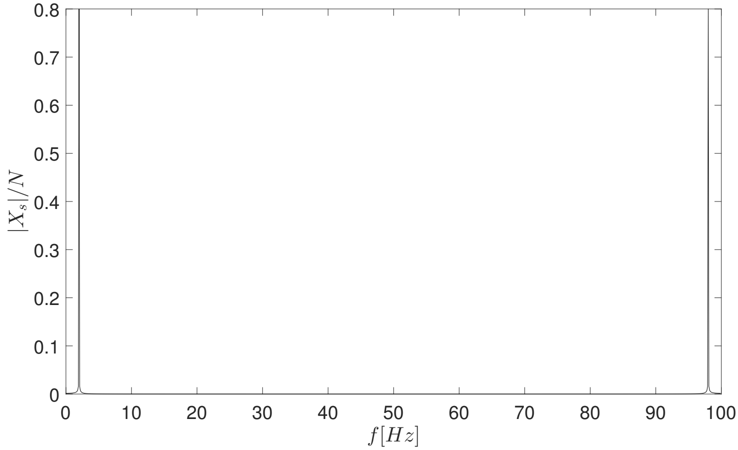
- Command 1: $Xs = fft(x);$
- Command 2: $N = size(x);$
- Command 3: $freq = [0:1:N-1]*fs/N;$

Step 2

We obtain the amplitude associated with each frequency component. This can be made by taking the absolute value of the complex numbers. Factor N is due to the algorithm.

- Command 4: $absXs = abs(Xs)/N;$

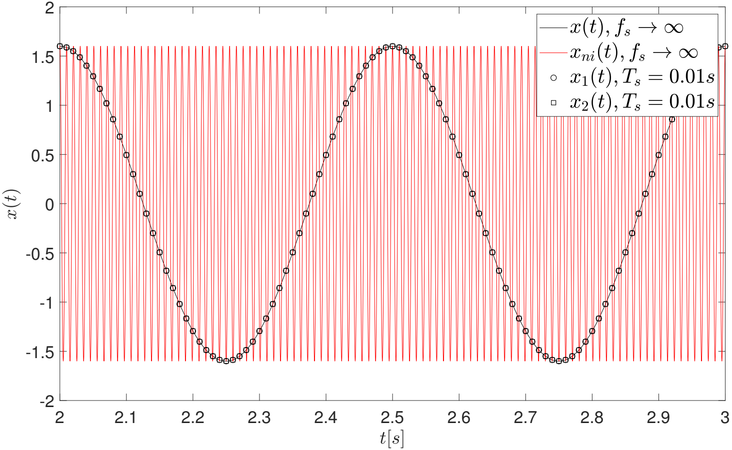
Result at the end of step 2



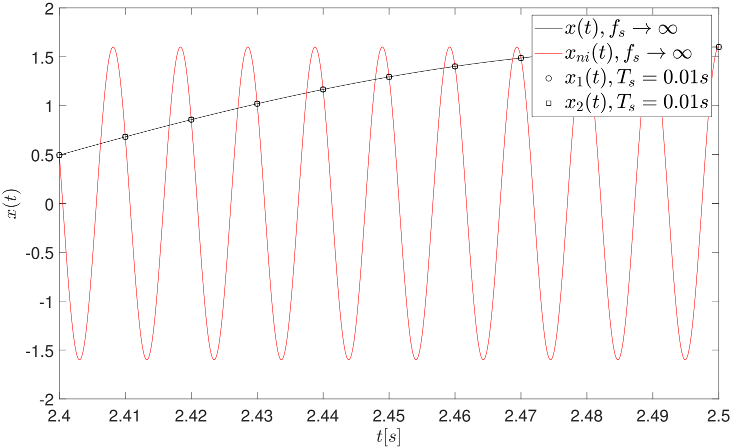
Step 3

The spectral analysis has identified two frequencies, $2\mathrm{Hz}$ and $98\mathrm{Hz}$, both with half the amplitude from the continuum time signal. Notice the symmetry with respect to $f_s/2$. We will see what happens....

Time-histories



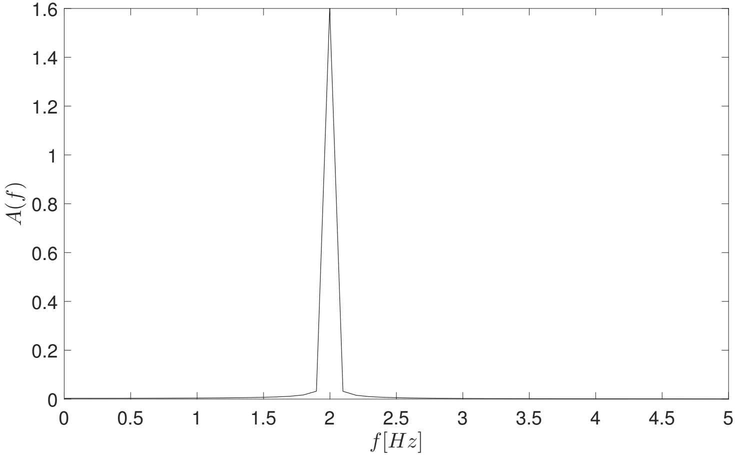
Time-histories



Amplitude spectrum $A(f)$

- We have two frequencies, the one of interest and another one of much larger value (*aliasing*). Each frequency is associated with half the amplitude of the original data;
- Aiming at obtaining a representative amplitude spectrum, we must consider only the frequencies below $f_s/2$. The amplitudes must be multiplied by 2;
- From this brief discussion, we can state that the largest frequency of interest of the signal sampled at f_s must be lower than $f_s/2$ (Sampling theorem or Nyquist theorem).

Amplitude spectrum $A(f)$



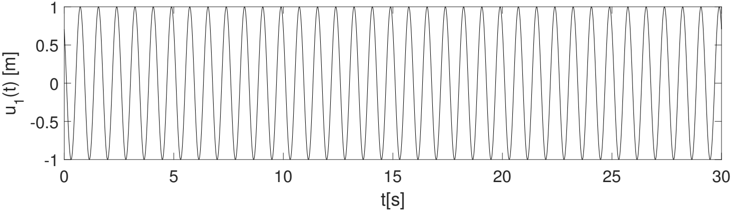
MATLAB® function

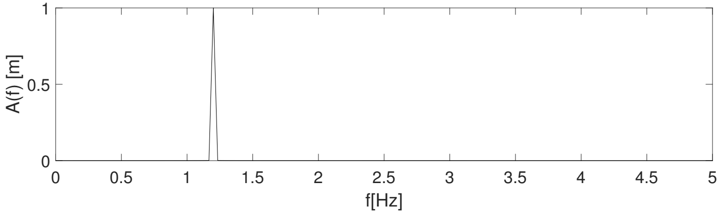
```
function [freq,Amp,fd,Ad]=EspectroAmplitude(t,sinal)

%% Calcula o espectro de amplitude do sinal
%[freq,Amp,fd,Ad]=EspectroAmplitude(t,sinal)
% Entradas - t -> vetor de tempo
%            sinal
% saida    - freq -> vetor de frequencias
%          - Amp  -> vetor de amplitudes
%          - fd   -> frequencia dominante
%          - Ad   -> amplitude na frequencia dominante
%%
N=length(t);
deltat=t(2)-t(1);
deltaf=1/deltat;
freq=[0:N-1]*deltaf/N;
Xs=fft(sinal);
Amp=abs(Xs)/N;
Amp=2*Amp(1:fix(N/2));
freq=freq(1:fix(N/2));
[Ad,indice]=max(Amp);
fd=freq(indice);
```

Outline

Ex. 1 - $u_1(t) = \rho\cos(2\pi f t + \theta)$

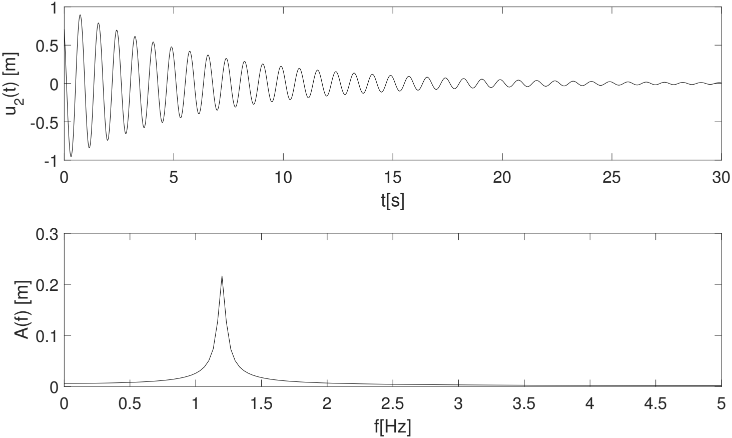
$$\rho = 1 \text{ m}; f = 1.2 \text{ Hz}; \theta = \pi/4$$





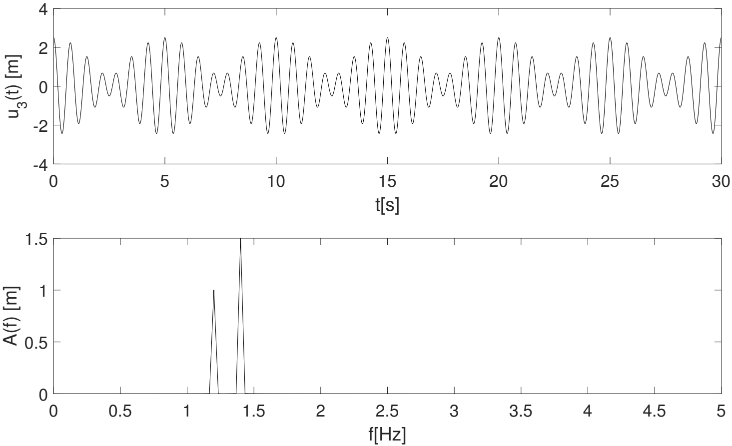
Ex. 2 - $u_2(t) = \rho e^{-\xi\omega t}\cos(2\pi f\sqrt{1-\xi^2}t+\theta)$

$$\rho = 1\ \mathrm{m}; f = 1.2\ \mathrm{Hz}; \xi = 0.02; \theta = \pi/4$$



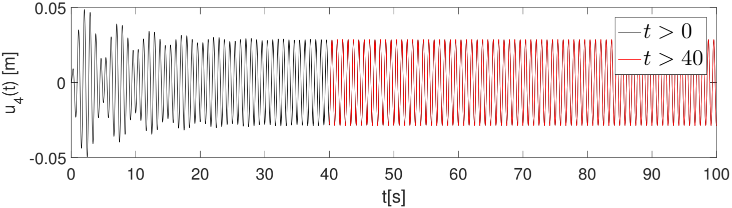
Ex. 3 - $u_3(t) = \rho_1 \cos(2\pi f_1 t) + \rho_2 \cos(2\pi f_2 t)$

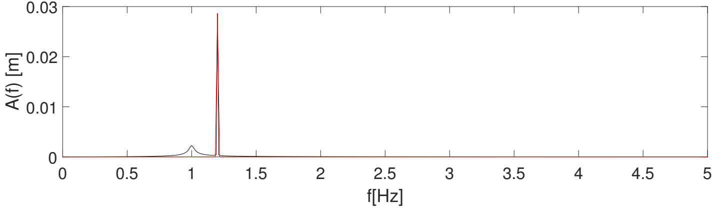
$$\rho_1 = 1\text{m};\, f_1 = 1.2 \text{ Hz};\, \rho_2 = 1.5\text{m};\, f_2 = 1.4 \text{ Hz}$$



Ex. 4 - Damped and forced 1-dof oscillator

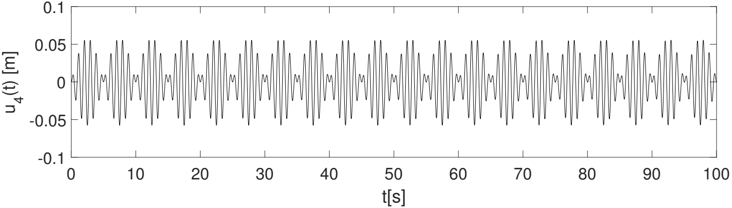
$$\ddot{u} + 2\zeta\omega\dot{u} + \omega^2 u = p_0/m\cos(\bar{\omega}t)$$
$$u(0) = 0 \text{ m}; \dot{u}(0) = 0 \text{ m/s}; m = 1 \text{ kg}; k = 4\pi^2 \text{ N/m}; \bar{\omega} = 1.2\omega; \xi = 2\%$$

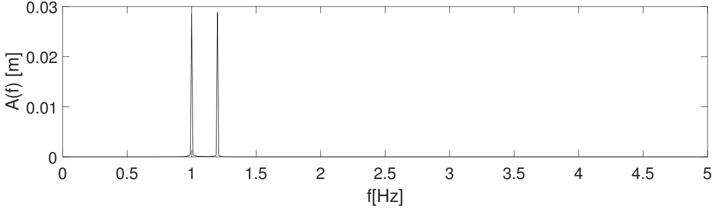




Ex. 5 - Undamped and forced 1-dof oscillator

$$\ddot{u} + 2\zeta\omega\dot{u} + \omega^2 u = p_0/m\cos(\bar{\omega}t)$$
$$u(0) = 0\ \mathrm{m}; \dot{u}(0) = 0\ \mathrm{m/s}; m = 1\ \mathrm{kg}; k = 4\pi^2\ \mathrm{N/m}; \bar{\omega} = 1.2\omega; \xi = 0$$





Outline

Definition

- For a stationary and ergodic process, the autocorrelation function of $f(t)$ is:

$$R_f(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) f(t+\tau) dt \tag{54}$$

- We define the power spectrum density as the Fourier Transform of the autocorrelation function

$$S_f(\omega) = \int_{-\infty}^{\infty} R_f(\tau) e^{-\hat{\imath}\omega\tau} d\tau \tag{55}$$

$$R_f(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_f(\omega) e^{\hat{\imath}\omega\tau} d\omega \tag{56}$$

- It can be proved that $R_f(\tau)$ is an even function of τ and $S_f(\omega)$ is an even function of ω.

Definition

- Using the above information, we have:

$$S_f(\omega) = \int_{-\infty}^{\infty} R_f(\tau) e^{-\mathbb{i}\omega\tau} d\tau = \int_{-\infty}^{\infty} R_f(\tau)(\cos\omega\tau - \mathbb{i}\sin\omega\tau) d\tau =$$

$$= \int_{-\infty}^{\infty} R_f(\tau)\cos\omega\tau d\tau = 2\int_{0}^{\infty} R_f(\tau)\cos\omega\tau d\tau \quad (57)$$

- Since $R_f(\tau)$ is a real function, Eq. 57 indicates that $S_f(\omega)$ is real;
- From the definition of $R_f(\tau)$ and recalling that $S_f(\omega)$ is an even function:

$$R_f(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_f(\omega) e^{\mathbb{i}\omega\tau} d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_f(\omega)(\cos\omega\tau + \mathbb{i}\sin\omega\tau) d\omega =$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} S_f(\omega)\cos\omega\tau d\omega = \frac{1}{\pi}\int_{0}^{\infty} S_f(\omega)\cos\omega\tau d\omega \quad (58)$$

- Equations 57 and 58 are known as Wiener-Khintchine relations.
- Important properties:

$$R_f(\tau = 0) = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} f^2(t) dt = \overline{f^2} \quad (59)$$

$$R_f(\tau = 0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_f(\omega) d\omega = \int_{-\infty}^{\infty} S_f(f) df = \overline{f^2} \quad (60)$$

Relation between $S_x(f)$ and $X(f)$

- Now, we discuss the relation between the Fourier Transform $X(f)$ and the PSD $S_x(f)$ of $x(t)$.
- We start from Parseval's theorem and from the definition of the mean square value of $x(t)$:

$$\int_{-\infty}^{\infty} x^2(t)dt = \int_{-\infty}^{\infty} X(f)X^*(f)df = \int_{-\infty}^{\infty} |X(f)|^2 df \qquad (61)$$

$$\overline{x^2} = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} x^2(t)dt = \lim_{T\to\infty} \frac{1}{T}\int_{-\infty}^{\infty} x^2(t)dt =$$

$$= \int_{-\infty}^{\infty} \lim_{T\to\infty} \frac{1}{T} X(f)X^*(f)df = \int_{-\infty}^{\infty} S_x(f)df \qquad (62)$$

- Hence

$$S_x(f) = \lim_{T\to\infty} \frac{1}{T} X(f)X^*(f) \qquad (63)$$

Practical aspects

- $p(t)$ is a real signal of null mean. Its Fourier series is

$$p(t) = \sum_{n=-\infty, n\neq 0}^{\infty} c_n e^{\mathbb{i} n\omega_0 t} \tag{64}$$

- This Fourier series can be given in terms of the positive frequency, as follows

$$p(t) = \sum_{n=1}^{\infty} \frac{1}{2}(c_n e^{\mathbb{i} n\omega_0 t} + c_n^* e^{-\mathbb{i} n\omega_0 t}) \tag{65}$$

- Now, we recall the orthogonality condition to obtain

$$\overline{p^2} = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} (p(t))^2 dt = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} \sum_{n=1}^{\infty} \frac{1}{4}(c_n e^{\mathbb{i} n\omega_0 t} + c_n^* e^{-\mathbb{i} n\omega_0 t})^2 dt \tag{66}$$

- Due to the orthogonality condition, we have

$$\overline{p^2} = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} \sum_{n=1}^{\infty} \frac{1}{4} 2 c_n c_n^* dt = \sum_{n=1}^{\infty} \frac{1}{2} c_n c_n^* \tag{67}$$

Practical aspects

- Notice that the terms c_n are the amplitude of the harmonic components;
- The power spectrum $G_n = G(f_n)$ is given by $\frac{1}{2}c_n c_n^*$;
- The power spectrum density $S_n = S(f)$ is given by $G_n/\Delta f$, Δf being the frequency interval;
- Units: If $p(t)$ represents the time-history of the applied force in N, $[G]$=N^2 and $[S]$=N^{2}s;
- If we would like to express the PSD considering the frequency in rad/s, we have:

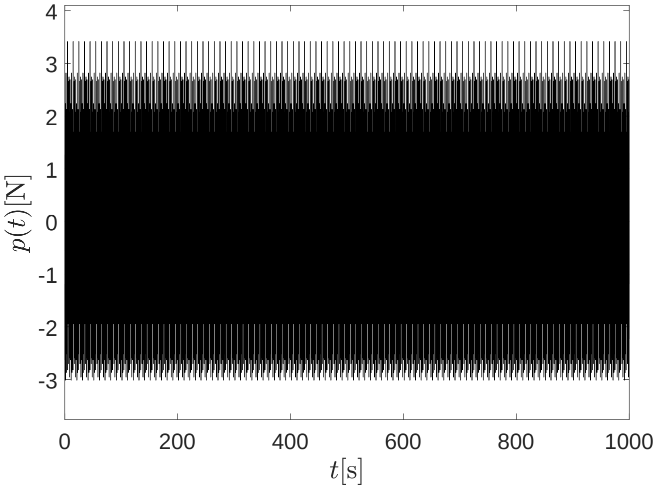
$$S(f)df = S(\omega)d\omega \leftrightarrow S(f)df = S(\omega)2\pi df \rightarrow S(f) = 2\pi S(\omega) \tag{68}$$

- Spectral moments of order k (m_k)

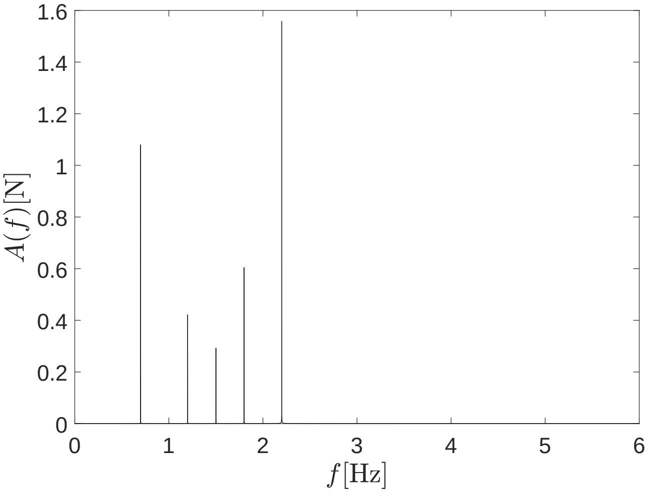
$$m_k = \int_0^\infty \omega^k S(\omega)d\omega \tag{69}$$

- Notice that the standard-deviation of the time-history is $\sqrt{m_0}$.

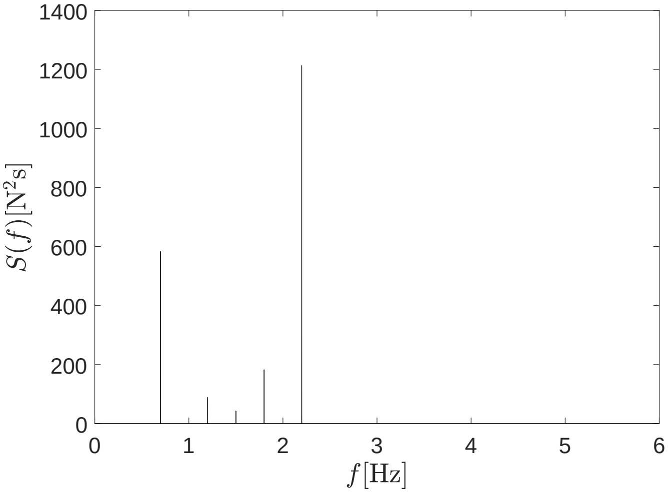
Example 1 : time history $p(t)$



Example 1 : Amplitude spectrum $A(f)$

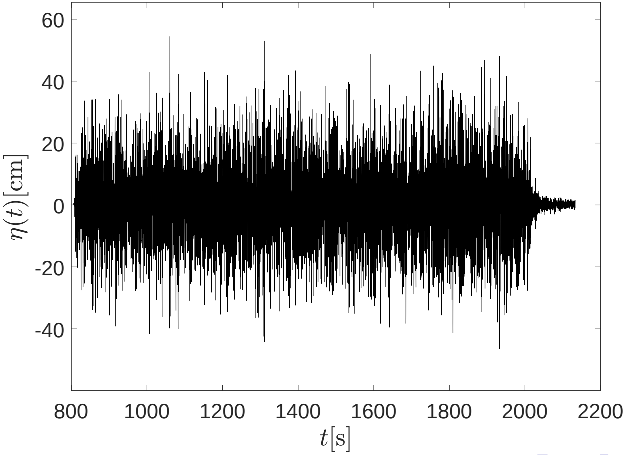


Example 1: PSD $S(f)$

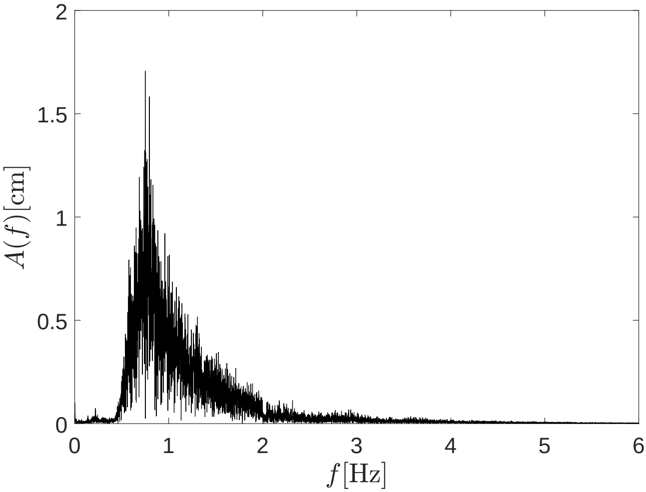


Example 2 : time-history $\eta(t)$

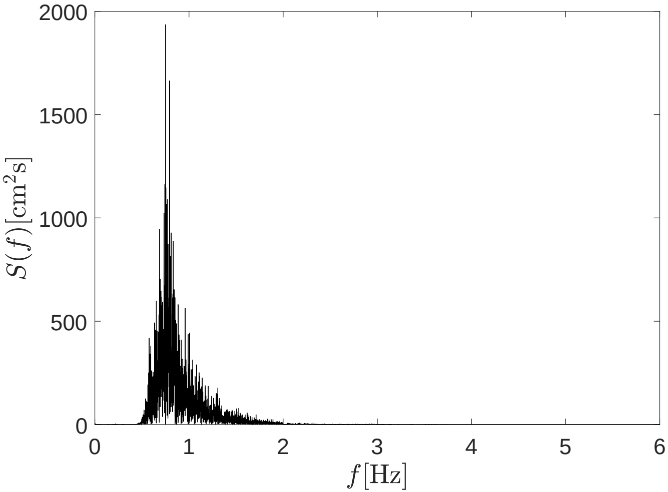
- $\eta(t)$ is the free-surface elevation time-history at a particular position. Data obtained from experiments carried out at TPN Wave Basin and kindly sent by PhD. Pedro Mello.
- The averaged value (offset) has been removed from data.



Example 2 : Amplitude spectrum $A(f)$



Example 2: PSD $S(f)$



- Standard deviation of $\eta(t)$ computed using std command: 12.4765
- Standard deviation of $\eta(t)$ computed using $\int_0^\infty S(f)df$: 12.4764

Outline

An important (and maybe, obvious) result

- We recall the Duhamel's integral

$$u(t) = \int_{-\infty}^{t} p(\tau)h(t-\tau)d\tau = \int_{0}^{\infty} p(t-\zeta)h(\zeta)d\zeta \tag{70}$$

- Since $h(\zeta) = 0, \zeta < 0$, we can write:

$$u(t) = \int_{-\infty}^{\infty} p(t-\zeta)h(\zeta)d\zeta \tag{71}$$

- Temporal average of $u(t)$:

$$\begin{aligned}\mathbb{E}[u(t)] = \mathbb{E}\left[\int_{-\infty}^{\infty} p(t-\zeta)h(\zeta)d\zeta\right] = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}\left[\int_{-\infty}^{\infty} p(t-\zeta)h(\zeta)d\zeta\right]dt = \\ = \int_{-\infty}^{\infty} h(\zeta)\left[\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} p(t-\zeta)dt\right]d\zeta = \int_{-\infty}^{\infty} h(\zeta)\mathbb{E}[p(t-\zeta)]d\zeta\end{aligned} \tag{72}$$

- Since the process is stationary, $\mathbb{E}[p(t-\zeta)] = \mathbb{E}[p(t)] = \mathrm{const}$ and

$$\mathbb{E}[u(t)] = \mathbb{E}[p(t)]\int_{-\infty}^{\infty} h(\xi)d\xi = \mathbb{E}[p(t)]H(0) \tag{73}$$

Another important result

- From the Duhamel's integral,

$$u(t) = \int_{-\infty}^{\infty} p(t-\zeta_1)h(\zeta_1)d\zeta_1, \;\; u(t+\tau) = \int_{-\infty}^{\infty} p(t+\tau-\zeta_2)h(\zeta_2)d\zeta_2 \tag{74}$$

- Autocorrelation function of $u(t)$: $R_u(\tau) = \mathbb{E}[u(t)u(t+\tau)]$

$$R_u(\tau) = \mathbb{E}[u(t)u(t+\tau)] = \mathbb{E}\left[\int_{-\infty}^{\infty} h(\zeta_1)p(t-\zeta_1)d\xi_1 \int_{-\infty}^{\infty} h(\zeta_2)p(t+\tau-\zeta_2)d\zeta_2\right] =$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(\zeta_1)h(\zeta_2)\mathbb{E}[p(t-\zeta_1)p(t+\tau-\zeta_2)]d\zeta_1 d\zeta_2 \tag{75}$$

- Since the process is stationary, $R_p(t,\tau,\zeta_1,\zeta_2) = \mathbb{E}[p(t)p(t+\tau-\zeta_2)]$ depends only on the time-shift $\tau+\zeta_1-\zeta_2$. Hence:

$$R_u(\tau) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(\zeta_1)h(\zeta_2)R_p(\tau+\zeta_1-\zeta_2)d\zeta_1 d\zeta_2 \tag{76}$$

Another important result

- From the definition of PSD:

$$S_u(\omega) = \int_{-\infty}^{\infty} R_u(\tau) e^{-\mathrm{i}\omega\tau} d\tau =$$
$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\zeta_1) h(\zeta_2) R_p(\tau + \zeta_1 - \zeta_2) d\zeta_1 d\zeta_2 e^{-\mathrm{i}\omega\tau} d\tau \tag{77}$$

- Also from its definition:

$$R_p(\tau + \zeta_1 - \zeta_2) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_p(\omega) e^{\mathrm{i}\omega(\tau + \zeta_1 - \zeta_2)} d\omega \tag{78}$$

- Using Eq. (78) in Eq. (77)

$$S_u(\omega) = \int_{-\infty}^{\infty} e^{-\mathrm{i}\omega\tau} \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\zeta_1) h(\zeta_2) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} S_p(\omega) e^{i\omega(\tau + \zeta_1 - \zeta_2)} d\omega \right] d\zeta_1 d\zeta_2 \right] d\tau$$
$$\int_{-\infty}^{\infty} e^{-\mathrm{i}\omega\tau} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} S_p(\omega) \left[\int_{-\infty}^{\infty} h(\zeta_1) e^{\mathrm{i}\omega\zeta_1} d\zeta_1 \int_{-\infty}^{\infty} h(\zeta_2) e^{-\mathrm{i}\omega\zeta_2} d\zeta_2 \right] e^{\mathrm{i}\omega\tau} d\omega \right] d\tau \tag{79}$$

Another important result

- Notice that $H(\omega) = \int_{-\infty}^{\infty} h(\zeta_1) e^{\mathrm{i}\omega\zeta_1} d\zeta_1$ and $H(-\omega) = \int_{-\infty}^{\infty} h(\zeta_2) e^{-\mathrm{i}\omega\zeta_2} d\zeta_2 = H^*(\omega)$
- Using the above results in Eq. (79), we have:

$$S_u(\omega) = \int_{-\infty}^{\infty} e^{-\mathrm{i}\omega\tau} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} S_p(\omega) \underbrace{H(\omega)H^*(\omega)}_{|H(\omega)|^2} e^{\mathrm{i}\omega\tau} d\omega \right] d\tau \tag{80}$$

- Recalling the definitions of the PSD and its inverse:

$$R_u(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_p(\omega) |H(\omega)|^2 e^{\mathrm{i}\omega\tau} d\omega \tag{81}$$

$$S_u(\omega) = S_p(\omega) |H(\omega)|^2 \tag{82}$$

- The operation given by Equation Eq. (82) is also know as spectral crossing;
- We can obtain the statistics of the response following the approach already developed;
- In some applications, the PSD of the excitation is given by empirical expressions. Hence, we can estimate statistics of the response without the numerical integration of the equations of motion in the time domain.

Other important aspects

- PSD is also obtained by $S_u(f) = \frac{A_u^2(f)}{2df}$
- Notice that we can obtain the amplitude of the harmonic components of velocity $\dot{u}$ as $2\pi f A(f)$. Then, the PSD of the velocity is written as

$$S_{\dot{u}}(f) = (2\pi f)^2 \frac{A_u^2(f)}{2df} = (2\pi f)^2 S_u(f) \tag{83}$$

- Using the same approach:

$$S_{\ddot{u}}(f) = (2\pi f)^2 S_{\dot{u}}(f) = (2\pi f)^4 S_u(f) \tag{84}$$

- If we have $S_u(f)$, we can obtain a realization of the time-history $u(t)$ by computing:

$$u(t) = \sum_{n=1}^{N} \sqrt{2S_u(f)df} \cos(2\pi f_n t + \phi_n) \tag{85}$$

ϕ_n being a random phase.

Practical example

Consider that the time history of excitation $p(t)$ has the PSD below illustrated with $\omega_0 = 1$ rad/s, $\omega_1 = 3$ rad/s, $\omega_2 = 6$ rad/s and $\hat{S} = 2$ N^{2}s. You must write a script for obtaining $p(t)$ from $S(\omega)$. The script must also obtain the response $u(t)$ of a 1-dof linear system of mass $m = 1$ kg, damping constant $c = 1$ Ns/m and stiffness $k = 10$ N/m excited by $p(t)$.